Toronto Crime and Demographic Analysis

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PROBLEM STATEMENT

Crime is a major concern in large metropolitan cities. Understanding the patterns and distribution of crime can help policymakers and law enforcement agencies allocate resources effectively. This project aims to analyze crime data across Toronto neighborhoods and examine potential relationships between crime levels and socioeconomic factors such as income, employment, and demographic characteristics.

Key Questions

- Which neighborhoods experience the highest crime levels?
- How do crime patterns change over time?
- Is there a relationship between crime and socioeconomic conditions?

OBJECTIVE AND STUDY

Project Objectives

- The primary objectives of this project are:
- To analyze crime incidents across Toronto neighborhoods.
- To identify trends in different types of crimes over time.
- To integrate demographic and socioeconomic data with crime statistics.
- To build a structured data model for analytical reporting.
- To visualize crime patterns using interactive dashboards.

By combining multiple datasets and analytical tools, this project aims to generate insights that can support data-driven decision making.

DATA SOURCE

This project integrates three datasets from the City of Toronto open data portal to analyze crime patterns and socioeconomic characteristics across neighborhoods.

Neighbourhood Profiles

This dataset provides demographic and socioeconomic statistics for each Toronto neighborhood

Neighbourhood Crime Rates Open Data

This dataset provides aggregated crime statistics by neighborhood and year.

Major Crime Indicators Open Data

This dataset contains detailed records of major criminal incidents reported by the Toronto Police Service.

ETL PROCESS OVERVIEW

01

Extract

Imported crime data and demographic data from external sources

02

Transform

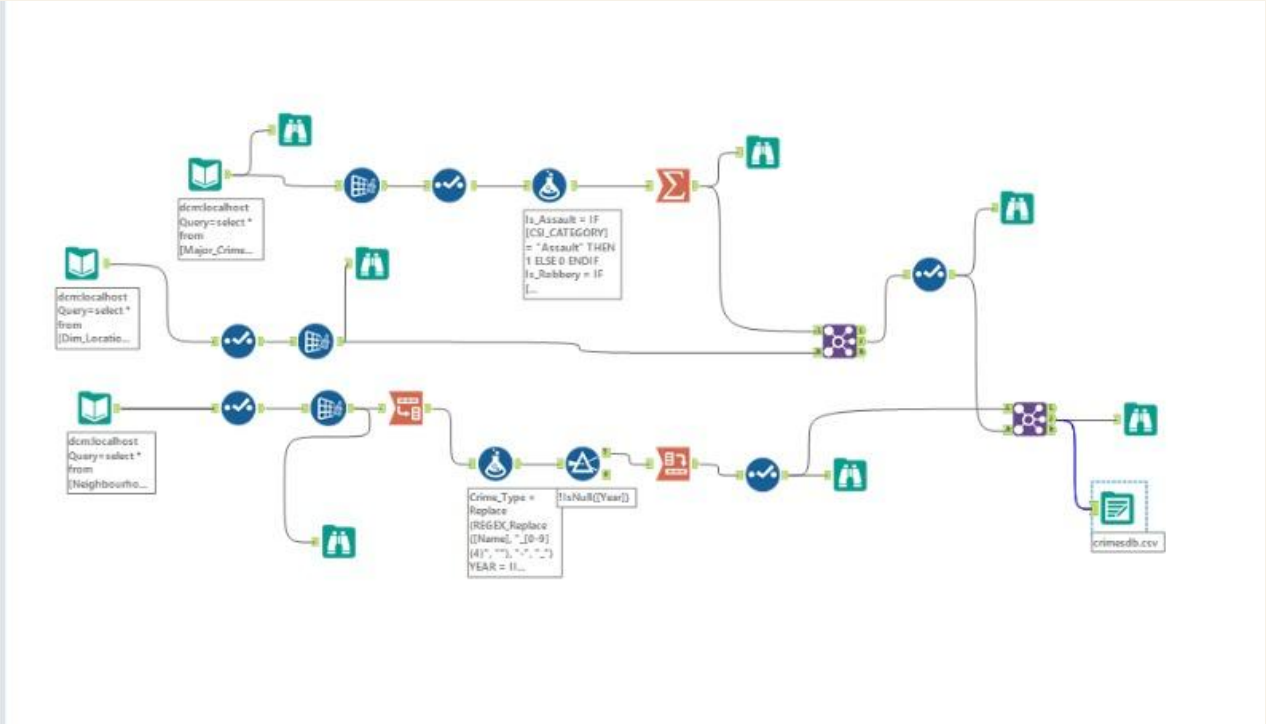
- Cleaned missing or null values.
- Standardized neighborhood naming conventions.
- Created binary indicators for crime types.
- Aggregated crime counts by neighborhood and year.
- Joined demographic data with crime data.

03

Load

Prepared the transformed dataset for use in the analytical data model.

ETL PROCESS OVERVIEW



DATA CLEANING & TRANSFORMATION

Several transformations were applied to ensure data quality and consistency:

- Removed incomplete or duplicate records.
- Replaced null numeric values with zero where appropriate.
- Standardized neighborhood naming formats to ensure accurate joins.
- Created indicator fields for specific crime categories such as:
  - Assault
  - Robbery
  - Auto Theft
  - Break and Enter
  - Theft Over

DATA AGGREGATION

The dataset was aggregated using grouping operations based on:

- Neighborhood name
- Report year

For each group, the total counts for each crime category were calculated.

Example measures generated:

- Total Assault Incidents
- Total Robbery Incidents
- Total Auto Theft Incidents
- Total Break and Enter Incidents
- Total Theft Over Incidents

Record	NEIGHBOURHOOD_NAME	POPULATION	YEAR	ASSAULT_COUNT	ASSAULT_RATE	AUTOTHEFT	AUTOTHEFT_RATE	BIKETHEFT	BIKETHEFT_RATE	BREAKENTER	BREAKENTER_RATE	HOMICIDE	HOMICIDE_RATE
1	Caledonia-Fairbank	10682	2025	59	552.330994	30	280.846283	2	18.723085	13	121.700058	0	0
2	Eglinton East	25914	2024	200	773.484924	69	266.852295	15	58.011372	35	135.359863	0	0
3	Yonge-Eglinton	14709	2022	122	908.211121	34	253.108017	14	104.220947	33	245.663666	1	7.444354
4	Islington	27977	2014	89	385.298065	28	121.217369	14	60.608685	54	233.776352	0	0
5	Downtown Yonge East	25119	2014	342	2587.185059	26	196.686584	158	1195.249268	88	665.708435	1	7.564869
6	Woodbine-Lumsden	8625	2025	58	672.463745	15	173.91304	2	23.188406	6	69.565216	1	11.594203
7	Danforth	10245	2019	96	959.13678	13	129.883102	28	279.74823	40	399.64032	0	0
8	Palmerston-Little Italy	14054	2024	69	482.990326	15	104.997902	49	342.993134	44	307.993835	0	0
9	Bendale South	11539	2023	73	627.902954	25	215.035263	2	17.202822	15	129.021164	0	0
10	Newtonbrook East	18231	2019	88	519.020935	33	194.632858	8	47.183723	53	312.592163	0	0
11	Greenwood-Coxwell	15496	2014	112	755.327759	16	107.903969	16	107.903969	48	323.711884	2	13.487996
12	Bendale South	11539	2016	78	706.777832	8	72.490036	7	63.42878	28	253.715118	1	9.061255

and more columns...

DATA MODEL

Star Schema Structure

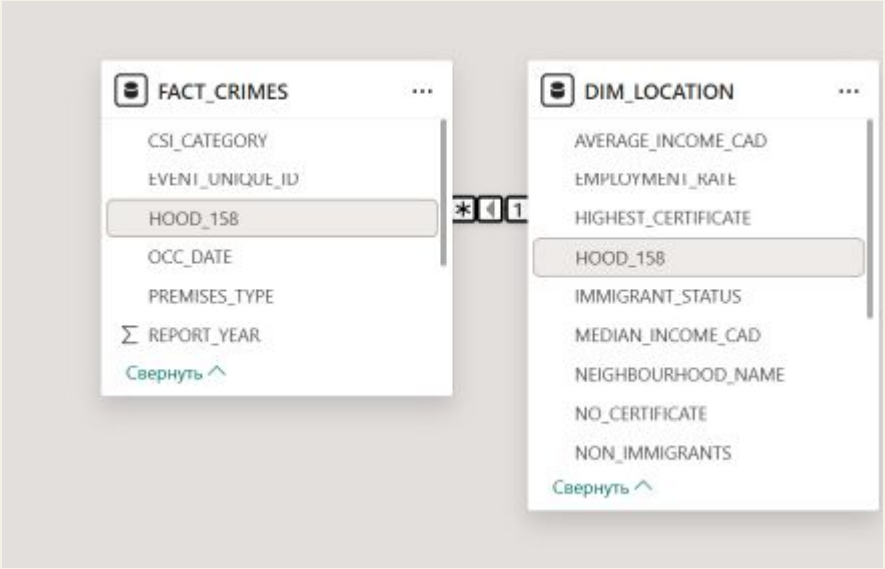
Fact Table – Fact_Crimes
Crime incident records
Fields: Event ID, Report Year, Crime Category, Premises Type, HOOD_158

Dimension Table – Dim_Location
Neighborhood demographic information
Fields: Neighborhood name, income, employment, housing data

```sql
-- 1. Creating the Dimension Table (Integrating Datasets 2 & 3)
CREATE OR REPLACE TABLE Dim_Location (
    NEIGHBOURHOOD_NAME VARCHAR(255),
    HOOD_158 INT PRIMARY KEY,
    VISIBLE_MINORITY INT,
    PRIVATE_HOUSEHOLDS INT,
    OWNER INT,
    RENTER INT,
    IMMIGRANT_STATUS INT,
    NON_IMMIGRANTS INT,
    PREVALENCE_LOW_INCOME FLOAT,
    UNEMPLOYMENT_RATE FLOAT,
    MEDIAN_INCOME_CAD FLOAT,
    AVERAGE_INCOME_CAD FLOAT,
    EMPLOYMENT_RATE FLOAT,
    HIGHEST_CERTIFICATE INT,
    NO_CERTIFICATE INT
);

-- 2. Creating the Fact Table (From Dataset 1)
CREATE OR REPLACE TABLE Fact_Crimes (
    EVENT_UNIQUE_ID VARCHAR(255) PRIMARY KEY,
    OCC_DATE DATE,
    REPORT_YEAR INT,
    CSI_CATEGORY VARCHAR(255),
    PREMISES_TYPE VARCHAR(255),
    HOOD_158 INT,
    FOREIGN KEY (HOOD_158) REFERENCES Dim_Location(HOOD_158)
);
```

DATA MODEL



Relationship

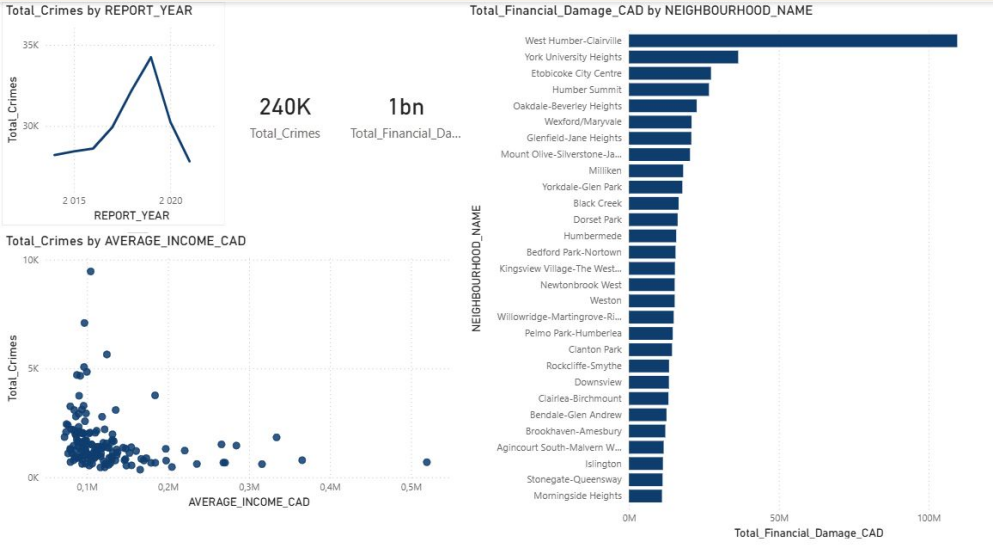
Dim_Location (1) → Fact_Crimes (*)

MEASURES & AGGREGATIONS

To test the cube and enable interactive reporting, the following calculated measures were defined:

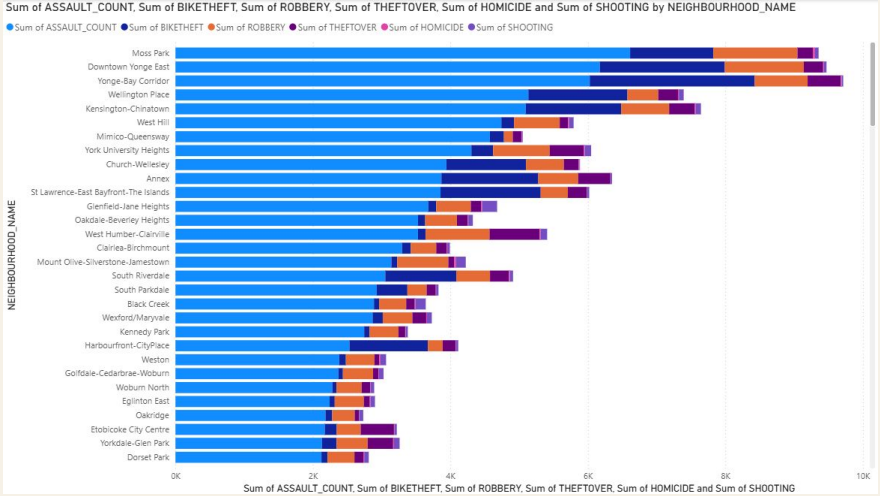
- Total Crimes = COUNT(Fact Crimes[EVENT UNIQUE ID])
- Total Auto Thefts = CALCULATE([Total Crimes], Fact Crimes[CSI CATEGORY] = "Auto Theft")
- Total Financial Damage CAD = [Total Auto Thefts] * 40000

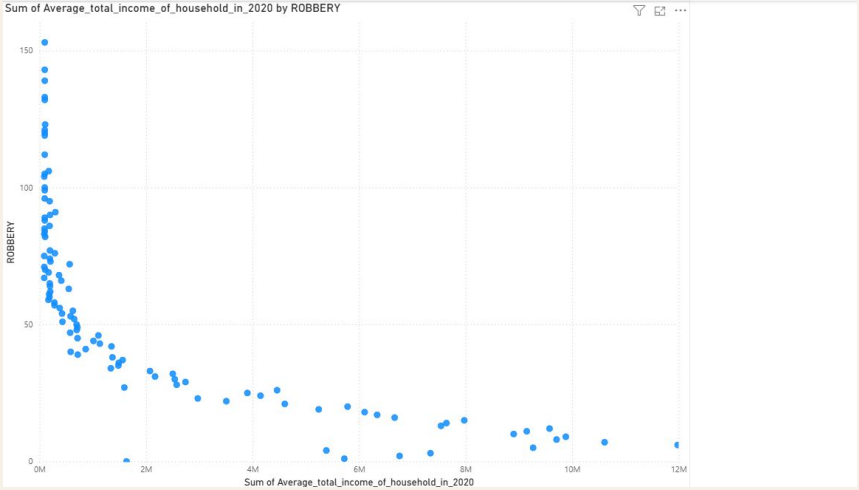
POWER BI DASHBOARD

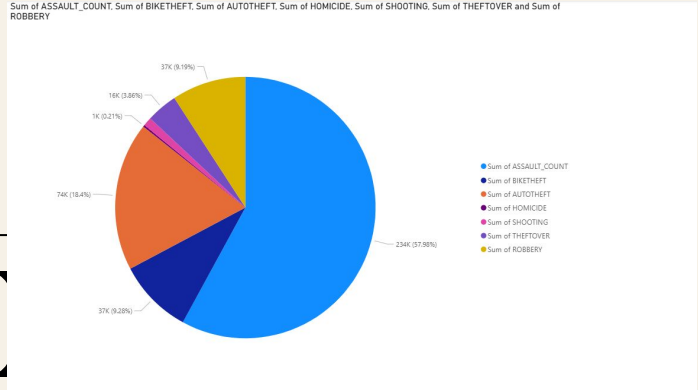


Main Interactive Dashboard: Overall Crime Trends, Wealth vs. Crime, and AutoTheft Damage.

POWER BI DASHBOARD

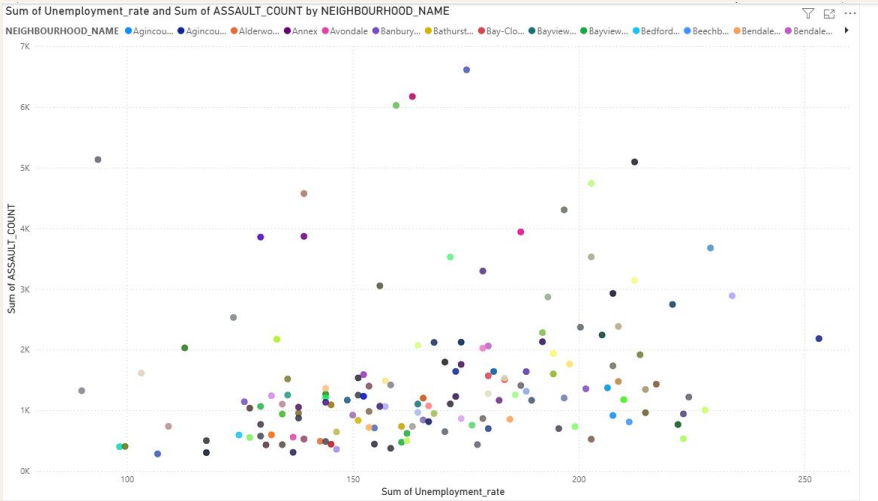


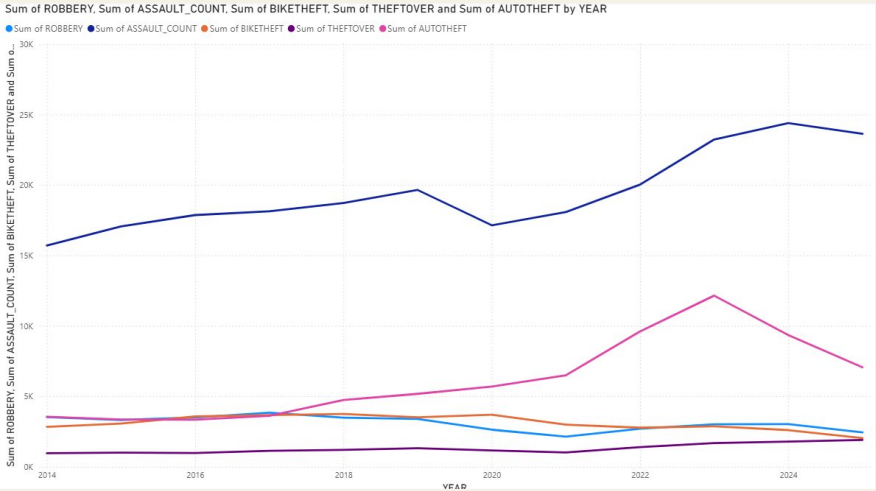




Insights:
Socioeconomic factors such as income levels may influence certain crime types, particularly robbery.
Crime is not evenly distributed across the city. High-density urban neighbourhoods tend to experience higher crime counts due to population concentration, nightlife activity, and economic factors.
Most crime in the dataset is concentrated in non-lethal but frequent offences, particularly assault and property-related crimes.

POWER BI DASHBOARD





Economic conditions such as unemployment may contribute to increased crime levels, but additional factors such as population density and urban activity also play a role.

Crime trends reveal that certain offences, particularly assault and auto theft, have increased in recent years. This may indicate changing crime patterns or external factors influencing crime rates during that period.

KEY INSIGHTS

Several important patterns emerged from the analysis:

- Assault is the most common major crime, representing the majority of incidents.
- Crime trends increased in recent years, particularly for auto theft.
- Downtown neighbourhoods experience significantly higher crime levels than suburban areas.
- Socioeconomic factors such as income and unemployment show potential correlations with crime levels.

These findings demonstrate how combining crime datasets with neighbourhood demographic information can help identify patterns and relationships that may support urban planning, law enforcement strategies, and community safety initiatives.

LIMITATIONS

While the analysis provides valuable insights, several limitations should be considered:

- Crime reporting may not capture all incidents due to underreporting.
- Demographic data represents aggregated neighborhood statistics and may not reflect individual circumstances.
- The analysis focuses primarily on historical data rather than predictive modeling.

Future studies could incorporate additional datasets or advanced analytical methods.

CONCLUSION

This project demonstrated how data analytics tools can be used to analyze complex urban datasets.

Key accomplishments include:

- Developing an ETL pipeline for integrating crime and demographic data.
- Building a structured tabular model for analytical queries.
- Creating interactive visualizations to explore crime patterns.
- Identifying relationships between crime and socioeconomic indicators.

Overall, the project highlights the value of data-driven analysis in understanding and addressing urban crime challenges.